



Cairo University
Faculty of Engineering
Credit Hours System



Project Proposal

CMPS344: Applied Data Science

Group 20

Group Members:

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Problem Statement:

Business Problem

Airlines and passengers suffer from unpredictable delays. While weather is a known factor, the interaction between weather severity and high-volume holiday periods is less understood. Our project aims to predict if a flight will be 'Significantly Delayed' (>15 minutes) based on historical performance, real-time climate conditions, and US holiday schedules.

Stakeholder

Booking platforms and passengers

Justification for classification

This project addresses the business problem through Supervised Binary Classification. We define our target variable as a Significant Delay (Binary: 1 if Delay > 15 mins, 0 otherwise).

Datasets Documentation:

We are working with 3 data sources.

1. US Domestic Flight Data (2024) (Primary Data source)

- Name: Flight Status Prediction 2024
 - Source URL: [Kaggle - Flight Data 2024](#)
 - Citation: Patil, H. (2024). *Flight Status Prediction Dataset*. Derived from the Bureau of Transportation Statistics (BTS).
 - Target Definition: Is_Delayed (Binary). A flight is classified as 1 (Delayed) if arr_delay > 15 minutes, otherwise 0 (On-Time).
 - Class Distribution:
- | is_delayed | Count | Percentage |
|------------|---------|------------|
| 0 | 5675693 | 80.18 |
| 1 | 1403388 | 19.82 |

2. Weather Data 2024 (Secondary Data source)

- Name: Weather Data 2024
- Source URL: [Kaggle - Weather Data 2024](#)
- Citation: Zain, S. (2024). *Global Weather Data 2024*.
- Target Definition: N/A (Provides explanatory features for the Is_Delayed target in the Primary set).

3. Global Airports Dataset (IATA, ICAO, Timezone)

- Name: Global Airports (IATA, ICAO, Timezone, Geo)
- Source URL: [Kaggle - Global Airports](#)
- Citation: Kocharyan, S. (2024). *Global Airports Database: IATA/ICAO Codes and Geolocation*.

- Target Definition: N/A (This is a Reference/Lookup Table, not a source for the target variable).
- Class Distribution: N/A (Static reference data for ~8,000+ global airports).

Features Description:

#	Feature	Source	Type
1	airline	Flight	Categorical
2	origin	Flight	Categorical
3	dest	Flight	Categorical
4	distance	Flight	Numerical
5	dep_hour	Engineered from departure_time in Flight	Numerical
6	day_of_week	Engineered	Categorical
7	month	Engineered	Categorical
8	is_weekend	Engineered	Binary
9	precipitation	Weather	Numerical
10	cloud_cover	Weather	Numerical
11	visibility	Weather	Numerical

Target Variable:

Variable	Derived From	Type	Definition
is_delayed	dep_delay > 15	Binary	1 = Delayed (>15 min), 0 = On time (≤15 min)

Data Validation Report:

1. Flight Delay Dataset
- a. Dataset file: flight_data_2024.csv

b. Rows: 7079081

c. Columns: 35

d. Duplicate rows: 0

e. Columns with >30% missing: cancellation_code

f. Columns with one unique value: year

g. Columns with leading/trailing spaces: None

h. Columns with inconsistent casing: None

i. Numeric columns with outliers: dep_delay, arr_delay, taxi_in, distance, crs_elapsed_time, taxi_out, air_time, actual_elapsed_time, op_carrier_fl_num

2. Weather Dataset

- a. Dataset file: weather_data.csv
- b. Rows: 1000000
- c. Columns: 6
- d. Duplicate rows: 0
- e. Columns with >30% missing: None
- f. Columns with one unique value: None
- g. Columns with leading/trailing spaces: None
- h. Columns with inconsistent casing: None
- i. Numeric columns with outliers: None

3. Airports Dataset

- a. Dataset file: airports.csv
- b. Rows: 6393
- c. Columns: 13
- d. Duplicate rows: 0
- e. Columns with >30% missing: None
- f. Columns with one unique value: None
- g. Columns with leading/trailing spaces: AirportName, ICAO, City_Name, City_IATA, Country_CodeA3
- h. Columns with inconsistent casing: AirportName, City_Name
- i. Numeric columns with outliers: None